

Sofia ML Regression — Solution Notes

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The [competition](#) is a tabular regression task: predict a continuous `target` from 20 anonymous numeric features (`x0..x19`), with 2500 rows to train on and 2500 to predict, scored by R².

TL;DR

Those 20 features turned out to relate to the target in a straight line, buried in noise with rare but huge outliers. So the final model is deliberately simple: fill missing values with the column median, fit plain linear regression, and drop five features that carry no signal (`x1`, `x5`, `x8`, `x9`, `x11`).

The real work was testing ideas fairly: every candidate was scored on the same cross-validation folds, so even tiny differences could be trusted. Judged that way, every fancier approach lost to plain linear regression, and dropping the five dead features was the only change that helped (CV R² 0.5348 to 0.5369, winning all 15 folds). The private score, 0.54051, landed almost exactly on that estimate — the validation was honest, and the model sits near the noise ceiling of this dataset.

EDA

Size	2500 train and 2500 test rows; 20 features (<code>x0..x19</code>) + <code>target</code> + <code>Id</code>
Missing values	about 1% of feature cells; the target column is complete
Features	roughly centered, very different scales
Correlations	features are essentially uncorrelated with each other (max 0.056)
Target	mean 1.8, std 149, range −1215 to 968

Fitting a straight-line model and studying what's left over tells most of the story: the leftover noise is large (std $\approx$ 101 against a target std of 149), perfectly symmetric, but far more outlier-prone than normal noise — the worst point sits eleven standard deviations out. Linear signal, heavy-tailed noise; the experiments below all make sense in that light.

EXPERIMENTS

In the order they happened:

WHAT WAS TRIED	WHAT HAPPENED
A fair test bench: 5-fold cross-validation repeated 3 times, same folds for every model, preprocessing refit inside each fold	tells apart models only $\sim 0.002 R^2$ apart; used for everything below
Baselines: median fill + linear regression, plus ridge and lasso (<code>cv_baselines.py</code>)	0.5348, all three essentially tied — regularization had nothing to fix
Curved fits: squared and interaction terms, gradient boosting	0.45-0.52, all clearly worse — the relationship really is a straight line
Outlier-resistant fitting (Huber loss)	~ 0.48 , worse — the outliers are symmetric noise, so ignoring them just throws away data
Smarter missing-value handling, weighted fitting, training only on complete rows	all matched or lost to the simple median fill
Dropping weak features step by step, in the order lasso ranked them (<code>cv_prune.py</code>)	dropping x1, x5, x8, x9, x11 gained +0.0021 and won all 15 folds — the final model

SUBMISSIONS

TAG	MODEL	CV R^2	PUBLIC R^2
<code>01_baseline_linear</code>	median fill + linear regression, all 20 features	0.5348	0.57121
<code>02_pruned_linear</code>	median fill + linear regression, 5 features dropped	0.5369	0.57186

CODE

Everything lives at github.com/HYLTE/ml-hw-summer-2026-python/tree/main/challenge.

The final score comes from `make_submission.py`: running `python make_submission.py 02_pruned_linear` rebuilds the uploaded file byte for byte. `cv_baselines.py` is the test bench and baseline comparison; `cv_prune.py` is the feature-dropping sweep that picked the final model.

To reproduce: `uv sync` at the repo root, put the competition CSVs in `challenge/data/` (not committed, per the data rules), then `uv run python challenge/<script>.py`.